

Maciej Robert Szczesny

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PROFESSIONAL EXPERIENCE

Student Researcher at Condensense (iGEM), Edinburgh, UK

MAR2026–PRESENT

- Working on biomolecular condensate-based biosensors for the iGEM competition as part of the Funding and Dry Lab teams.
- Molecular Dynamics using AMBER/OpenMM, testing variations of our construct in silico, and visualising results and presenting them to the wet lab team.
- Reaching out to potential sponsors and industry partners, secured Eppendorf and Vazyme Biotech funding.

Student Researcher at The University of Edinburgh, Edinburgh, UK

MAY2026–AUG2026

- Built *flashEE* from scratch in PyTorch: a 4.94M-parameter protein language model trained on eukaryotic UniProt sequences, designed to study the impacts of quantisation.
- Covered the full pipeline: data processing, model architecture, training and evaluation. Code at github.com/skurl/flashEE.
- Project funded by CCPBioSim and EPSRC.

Intern at the Centre of New Technologies, Warsaw, Poland

JUN2024–AUG2024

- Performed 5' cap synthesis by solid-state methods, verified by HPLC, TLC and mass spectrometry.
- Modifying mRNA for pharmaceutical purposes using techniques such as plasmid linearisation, IVT, cleanup using Monarch Kit, precipitation, freeze drying, nanodrop, dot plot to check for impurities, electrophoresis (agarose and polyacrylamide gel, SDS-PAGE), and isolation using 5' exonuclease RNase.
- Protein purification using the *E. coli* protein expression system, FPLC, MST, dialysis, centrifuge and a sonicator.
- Exposure to research financing, public versus private investment, and publishing.

Data Manager for Pharmaceutical Companies, Remote

JAN2020–PRESENT

- Clinical data entry for Regeneron, Roche and others, part-time job alongside full-time study.
- Good Clinical Practice (GCP) training completed.

Wet Lab Experience

- **Warsaw Genomics** (Jul–Sep 2022) — designed primers and ran PCR assays following standard laboratory protocol.
- **Dept. of Pathology and Laboratory Diagnostics, Warsaw** (Aug 2021) — microscopy and sectioning of paraffin-embedded tissues.

PROJECTS & EXTRACURRICULAR

Student Researcher at the Edinburgh Scientific Research Association

OCT2024–MAY2025

- Designed a project on plastic degradation by *Aspergillus niger* and outlined the assessment methods.
- Wrote the proposal and secured the funding.

Junior Analyst at the Edinburgh University Trading and Investment Club JAN2025–MAR2025

- Researched and pitched healthcare-sector investment proposals to the club board.

Volunteer at Hospicjum Stacjonarne im. św. Patryka, Poland

SEP2019–FEB2020

EDUCATION

The University of Edinburgh, Edinburgh

2023–PRESENT

- Fourth-year Undergraduate Biological Sciences (Biotechnology) Student, Current Avg. ~74%

St. Edward's, Poole

2020–2022

- Biology, Chemistry and Maths A-Levels

Copernicus High School, Warsaw

2019–2020

Taipei Jingwen Bilingual School, Taipei

2018–2019

Staszic Middle School, Warsaw

2016–2018

- Mathematics-focused Middle School with tailored advanced mathematics curriculum.

SKILLS

Programming: Python (PyTorch, NumPy, pandas), Bash, $\text{\LaTeX}$

Molecular dynamics: AMBER, OpenMM

Tools: Linux, Git, Microsoft Office Suite, Fusion360

Languages: Polish (Native), English (Fluent), Chinese (Upper-Intermediate), Russian (Emerging)

Polish citizen. UK Settled Status – Indefinite Leave to Remain (ILR)